# Year 11 Mathematics

## Graphs 6

8.2 Using Graphs to Solve Other Equations

# Using Graphs to Solve Other Equations

STARTER · 5 MIN

**Date:** _____ **Title:** Using Graphs to Solve Other Equations

**LEARNING OBJECTIVES**

- Recognise and draw cubic and reciprocal graphs
- Solve equations where a curve meets a straight line
- Find the line to draw to solve an equation

**STARTER** *Answer in your book. No calculators.*

1. Find y when x = 2 for $y = x^3$
2. Find y when x = −2 for $y = x^3$
3. Find y when x = 4 for $y = \frac{8}{x}$
4. Rearrange $x^2 = x + 2$ so that one side is 0

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. $2^3 = \mathbf{8}$

2. $(-2)^3 = \mathbf{-8}$

3. $8 \div 4 = \mathbf{2}$

4. $x^2 - x - 2 = \mathbf{0}$

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can recognise and draw cubic and reciprocal graphs | ☐ |
| SILVER | I can solve equations where a curve meets a straight line | ☐ |
| GOLD | I can find the line to draw to solve an equation | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Curves and Lines

INTRODUCTION · 3 MIN

Where a curve and a line cross, both equations are true. Read the x-values.

## Cubic

$y = x^3$ has an S-shaped curve.

Can cross the x-axis 3 times

## Reciprocal

$y = \frac{k}{x}$ has two separate branches.

Never touches the axes

## Curve and line

The solutions are the x-values where they cross.

Read every crossing point

## Find the line

Rearrange so one side is the graph you have.

The other side is the line

# Examples 1 and 2 — Where a Curve Meets a Line

EXAMPLES · 5 MIN

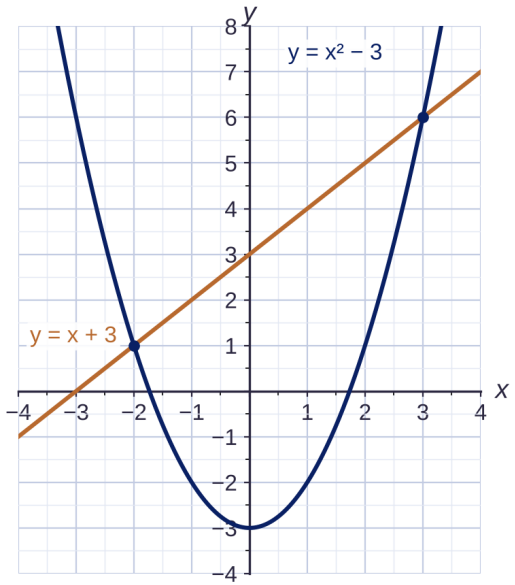
BRONZE

*Watch and copy into your book*

**Hint** solutions: x-values where the graphs cross

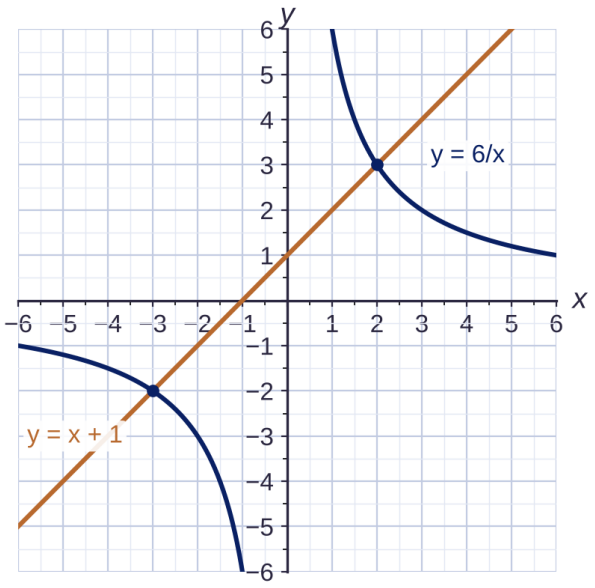
## Example 1

Use the graph to solve $x^2 - 3 = x + 3$.




## Example 2

Use the graph to solve $\frac{6}{x} = x + 1$.



# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

Read where they cross

$x = -2$

**$x = 3$**

Tip: give only the x-values.

## Example 2

The line meets both branches

$x = -3$

**$x = 2$**

Tip: check both branches.

# Task 1

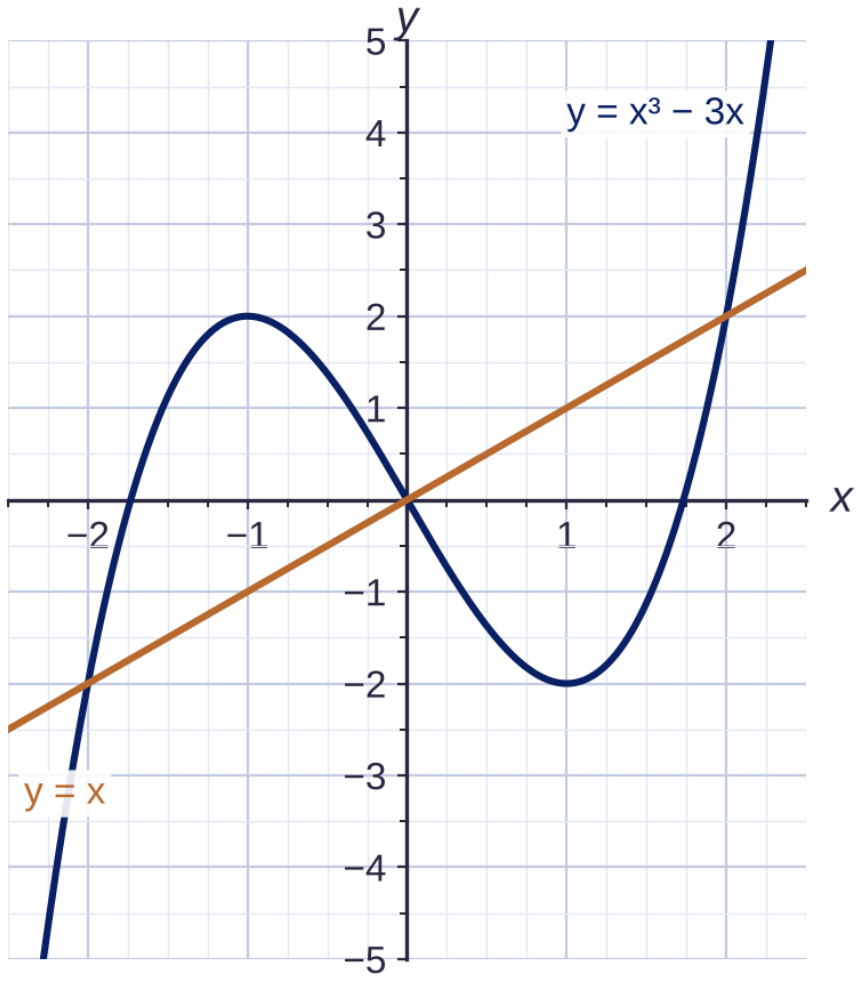
PRACTICE · 9 MIN

BRONZE *Use the graph of $y = x^3 - 3x$*

**Hint** solutions: x-values where the graphs cross

1. Solve $x^3 - 3x = 0$
2. Solve $x^3 - 3x = 2$
3. Solve $x^3 - 3x = x$
4. Solve $x^3 - 3x = -2$
5. Write down the turning points

★ How many solutions does $x^3 - 3x = 1$ have?



# Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. $x \approx -1.7$, $x = 0$, **$x \approx 1.7$**

2. $x = -1$ and $x =$ **2**

3. $x = -2$, $x = 0$, $x =$ **2**

4. $x = -2$ and $x =$ **1**

5. **(−1, 2) and (1, −2)**

★ Draw $y = 1$, **3 solutions**

# Examples 3 and 4 — Finding the Line to Draw

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

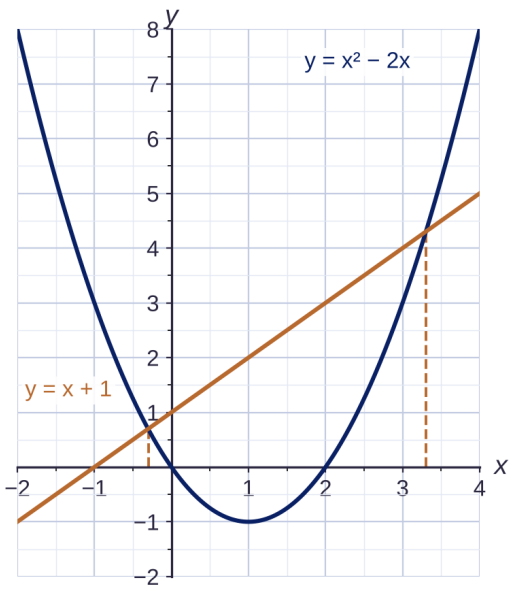
**Hint** rearrange so one side is the graph's equation

## Example 3

The graph of $y = x^2 - 2x$ is drawn. Which line must be drawn to solve $x^2 - 3x - 1 = 0$?

## Example 4

Use the graph to solve $x^2 - 3x - 1 = 0$, to 1 d.p.



# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

$x^2 - 3x - 1 = 0$

$x^2 - 2x = x + 1$

Draw y **= x + 1**

Tip: make one side match the graph.

## Example 4

Read where they cross

**x ≈ −0.3 and x ≈ 3.3**

Tip: read the x-values only.

# Task 2

PRACTICE · 9 MIN

SILVER

*Write the line to draw on the graph of $y = x^2 - 2x$*

**Hint** rearrange so one side is the graph's equation

1. To solve $x^2 - 2x - 5 = 0$
2. To solve $x^2 - 4x = 0$
3. To solve $x^2 - x - 3 = 0$
4. To solve $x^2 - 5x + 2 = 0$
5. To solve $x^2 = 3$

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. $x^2 - 2x = 5, \quad y = \mathbf{5}$

2. $x^2 - 2x = 2x, \quad y = \mathbf{2x}$

3. $x^2 - 2x = 3 - x, \quad y = \mathbf{3 - x}$

4. $x^2 - 2x = 3x - 2, \quad y = \mathbf{3x - 2}$

5. $x^2 - 2x = 3 - 2x, \quad y = \mathbf{3 - 2x}$

# Examples 5 and 6 — Reciprocal and Cubic Problems

EXAMPLES · 5 MIN

GOLD

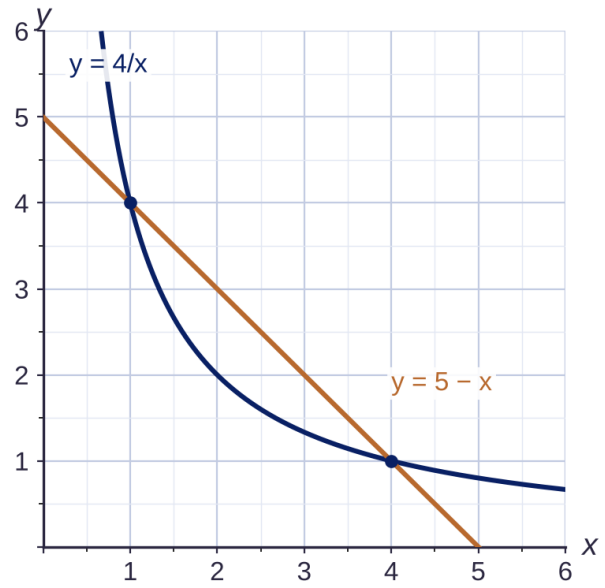
*Exam-style: show all your working*

**Hint** rearrange so one side is the graph's equation

## Example 5

(3 marks)

Use the graph to solve $\frac{4}{x} = 5 - x$.




## Example 6

(2 marks)

The graph of $y = x^3 - 2x^2$ is drawn. Find the line to draw to solve $x^3 - 2x^2 - x + 1 = 0$.

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

Crossing points (1, 4) and (4, 1)

$x = 1$

**$x = 4$**

Tip: check: $4 \div 1 = 5 - 1$.

## Example 6

$x^3 - 2x^2 = x - 1$

Draw $y$ **$= x - 1$**

Tip: move everything except $x^3 - 2x^2$.

# Task 3

PRACTICE · 9 MIN

GOLD

*Exam-style: show all your working. Give answers to 1 d.p.*

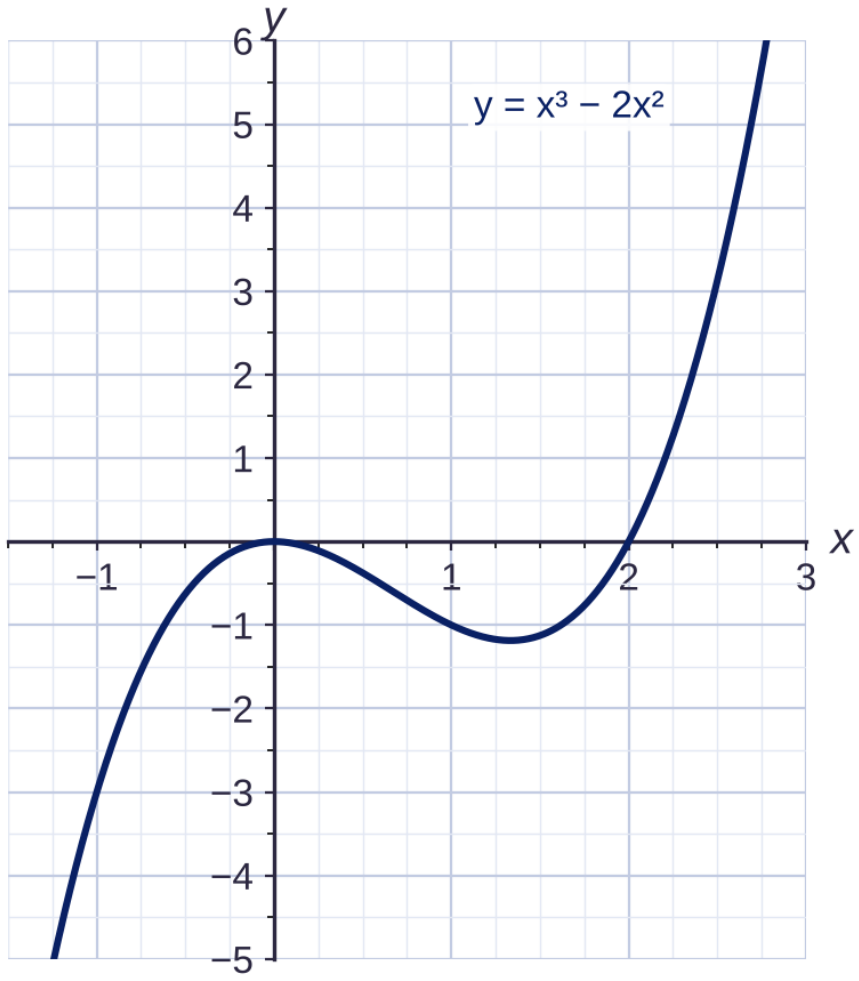
1. Solve $x^3 - 2x^2 = 0$ (1 mark)

2. Solve $x^3 - 2x^2 = -1$ (2 marks)

3. Which line should be drawn to solve $x^3 - 2x^2 - x = 0$? (2 marks)

4. Use your line to solve $x^3 - 2x^2 - x = 0$ (2 marks)

5. For which values of k does $x^3 - 2x^2 = k$ have three solutions? (2 marks)



# Task 3 — Answers

CORRECTIONS

GOLD

*Tick or correct in green pen*

1. $x = 0$ and $x = \mathbf{2}$

2. $x \approx -0.6$, $x = 1$, $\mathbf{x \approx 1.6}$

3. $x^3 - 2x^2 = x$, $y = \mathbf{x}$

4. $x \approx -0.4$, $x = 0$, $\mathbf{x \approx 2.4}$

5. $\mathbf{-1.2 < k < 0}$

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Where do $y = x^2$ and $y = 4$ cross?

**SILVER** Using $y = x^2 - 2x$, which line solves $x^2 - 2x - 3 = 0$?

**GOLD** Using $y = x^3$, which line solves $x^3 - x - 1 = 0$?

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

**BRONZE** $x^2 = 4, \quad x = -2$ and $x = \mathbf{2}$

**SILVER** $y = \mathbf{3}$

**GOLD** $x^3 = x + 1, \quad y = \mathbf{x} + \mathbf{1}$

# Exam Practice — Where a Curve Meets a Line

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

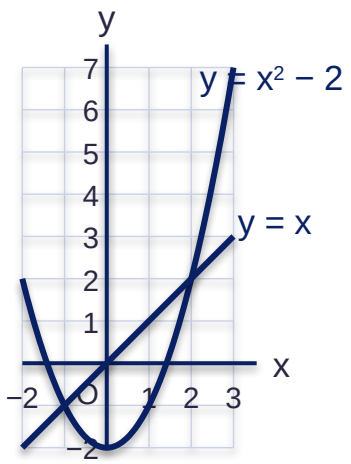
**Hint** solutions: x-values where the graphs cross

## Question 1

(2 marks)

The graphs of $y = x^2 - 2$ and $y = x$ are drawn.

Use them to solve $x^2 - 2 = x$.




x = .....

## Question 2

(2 marks)

The graph of $y = x^2$ is drawn.

Write down the equation of the line you would draw to solve $x^2 - x - 2 = 0$.

y = .....

# Exam Practice — Finding the Line to Draw

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint** rearrange so one side is the graph's equation

## Question 3

(2 marks)

The graph of $y = x^2 - 3x$ is drawn.

Write down the equation of the line you would draw to solve $x^2 - 5x + 2 = 0$.

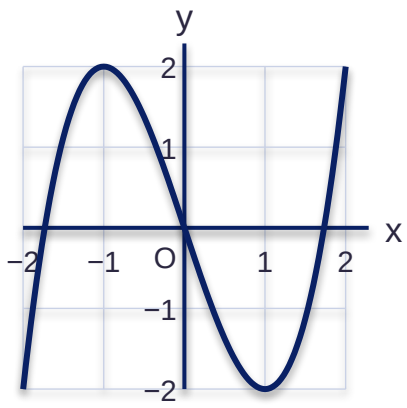
y = .....

## Question 4

(2 marks)

The graph of $y = x^3 - 3x$ is drawn.

How many solutions does $x^3 - 3x = 1$ have?




.....

# Exam Practice — Cubic Problems

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** rearrange so one side is the graph's equation

## Question 5

(2 marks)

The graph of $y = x^3 - 2x^2$ is drawn.

Find the line to draw to solve $x^3 - 2x^2 - 2x - 1 = 0$.

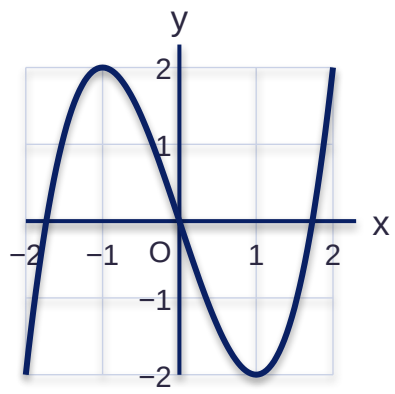
y = .....

## Question 6

(2 marks)

The graph of $y = x^3 - 3x$ is drawn.

For which values of k does $x^3 - 3x = k$ have three solutions?




.....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

read where the graphs cross — M1

**x = −1,  x = 2** — A1

Tip: only the x-values are needed.

## Question 2

$x^2 = x + 2$ — M1

**y = x + 2** — A1

Tip: make one side $x^2$.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

$x^2 - 3x = 2x - 2$ **M1**

**$y = 2x - 2$** **A1**

Tip: add 2x − 2 to both sides.

## Question 4

draw the line y = 1 **M1**

**3 solutions** **A1**

Tip: count the crossing points.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$x^3 - 2x^2 = 2x + 1$ **M1**

**$y = 2x + 1$** **A1**

Tip: keep $x^3 - 2x^2$ on one side.

## Question 6

turning points at $y = 2$ **M1**

and $y = -2$

**$-2 < k < 2$** **A1**

Tip: the line $y = k$ must cut all three parts.